

TORIN NELSON

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RESEARCH INTERESTS

Computational complexity, particularly randomness, derandomization, and the structural foundations of efficient computation; and theoretical machine learning, with particular interest in optimization, learning dynamics, and the computational requirements of learning systems.

EDUCATION

University of Colorado Boulder - Boulder, CO

M.S. in Computer Science (AI Specialization)

Expected May 2027

- GPA: 4.00 / 4.00 (21 completed graduate credits)

University of California, Irvine - Irvine, CA

B.S. in Computer Science (Specialization in Intelligent Systems)

June 2024

- GPA: 3.697 / 4.00

Las Positas College - Livermore, CA

A.S. in Computer Science & A.S.-T in Mathematics

May 2022

- GPA: 3.83 / 4.00

HONORS & AWARDS

- **Dean's Honor List**, UC Irvine - Awarded 5 quarters: Winter 2023, Spring 2023, Fall 2023, Winter 2024, Spring 2024.
- **Future Prospects Award**, UC Irvine CS 178 (Machine Learning) - Course-level recognition for the Attention U-Net satellite imagery remote sensing project (Fall 2023).

TECHNICAL & ENGINEERING EXPERIENCE

Sankaku Analytics - Remote

Founding Engineer

Feb 2026 - May 2026

- Developed the MVP for an early-stage retail analytics startup project, building high-performance data processing pipelines and backend systems.
- Implemented and optimized K-means clustering algorithms for automated retail data segmentation and pattern discovery.
- Architected system backends and data architecture from the ground up, making technical decisions on software stack selection and data integrity.
- Analyzed the client's proprietary retail data and presented cluster analysis findings and actionable technical insights directly to the client's CEO, Head of Retail, and Data Analyst.

NASA L'SPACE Mission Concept Academy - Remote

Deputy Project Manager & Command and Data Handling (CDH) Systems Lead

Apr 2024 - Sep 2024

- Co-led a 12-person student engineering team simulating NASA Phases A and B for a lunar mission concept to analyze rock-ice samples in lunar Permanently Shadowed Regions (PSRs).
- Co-authored and delivered a 240-page Preliminary Design Review (PDR) and led mission scope revisions to fit a simulated 22% budget reduction.
- Defined CDH subsystem requirements and variable data-transmission rates (30s science payload / 1s telemetry) for critical mission phases.
- Conducted trade studies for avionics and mechanical components (IMUs, motors), selecting hardware based on thermal operating ranges (-193°C to 111°C) and power constraints.

Morgan Stanley - Pleasanton, CA

Summer Intern

Jun 2018 - Aug 2018

- Analyzed market trends, financial and ESG information, and client and IRA-related data.
- Prepared reports and data presentations, assisted with office organization, and participated in professional and networking activities.
- Organized client data into a readily available physical resource to streamline client interactions.

SELECTED TECHNICAL & ACADEMIC PROJECTS

Computational Tone-Mapping (HDR-to-SDR Video Conversion Software)

Independent Project

Jan 2024 - Present

- Sole developer of a Windows desktop application for HDR-to-SDR video conversion and mathematical tone mapping.
- Designed a ground-up HDR-to-SDR color-processing pipeline implementing BT.2020 to BT.709 gamut mapping via a custom-generated 3D LUT with tetrahedral interpolation.
- Built GPU video processing pipelines using Vulkan and libplacebo with robust CPU fallback paths.
- Integrated automatic hardware encoder detection and acceleration for NVIDIA NVENC, AMD AMF, and Intel QSV.
- Engineered a thread-safe, asynchronous processing architecture with real-time tracking and rendering control.
- Developed offline hardware-locked licensing using HMAC-signed node-locked tokens, alongside automated installer, auto-updater, and code-signing release infrastructure.
- Maintained a 19-module automated test suite using unittest.mock to enforce a 90% code coverage floor across core processing paths.
- Shipped 26 production releases, with the application surpassing 2,500 downloads.

Attention U-Net Remote Sensing & Infrastructure Mapping

UC Irvine ML Course Project

Sep 2023 - Dec 2023

- Developed an Attention U-Net model using publicly available satellite imagery to estimate electricity infrastructure needs and map underserved regions in Sub-Saharan Africa.
- Achieved 75% segmentation accuracy on satellite remote sensing data.
- Applied explainable AI (XAI) techniques downstream of the prediction pipeline to interpret and visualize spatial feature representations influencing predictions.
- Received course-level "Future Prospects Award" in CS 178 (Machine Learning).

Quantitative Analysis & Statistical Signal Processing

Independent Project

Nov 2025 - Jan 2026

- Built an automated real-time statistical analysis system in Python for ingesting, scoring, and filtering high-frequency probability signals.
- Implemented a historical variance filtering model to analyze probability distributions and eliminate non-systematic noise signals.
- Developed a multithreaded polling architecture with variance estimation, slippage estimation, and consensus weighting for signal confidence scoring.

RELEVANT COURSEWORK

- **Graduate (University of Colorado Boulder):** Machine Learning, Deep Learning, Natural Language Processing, Reinforcement Learning, Algorithms, Autonomous Systems, Software Architecture, Network Systems
- **Undergraduate (University of California, Irvine):** Machine Learning and Data Mining, Formal Languages and Automata, Embedded Software, Algorithms and Data Structures

TECHNICAL SKILLS

- **Languages:** Python, C, C++
- **Machine Learning & Data:** PyTorch, scikit-learn, NumPy, pandas, Machine Learning, Deep Learning, NLP, Quantitative Analysis, Scientific Computing
- **Systems & Engineering Tools:** Vulkan, libplacebo, FFmpeg, Hardware Acceleration (NVENC, AMF, QSV), Git, Linux, REST APIs, Test-Driven Development, Technical Documentation